

INF-2900: SOFTWARE ENGINEERING

USER STORIES, SCENARIOS, PERSONAS AND FEATURES

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Personas

Personas 1: Business owner Ola is a business owner that is 35 year old. He is just finished with his masters degree in marketing from the Arctic University in Tromsø. He is originally from Sortland in Northern-Norway. He is engaged to Frida and they are expecting their first child in July 2026. Ola have recently been taken over the operation of the family business that is a bookstore. Ola have worked for some years worked in the business and via that found interest of marketing, and decided to formalize it. During his work he have seen that especially the inventory needs to modernized using technology that for the previous generation was not very appealing. Ola believes that a easily accessible digital inventory system will be of great value to both him as a owner and the employees in the business.

Personas 2: Business owner 2 Jessica is 21 years old and a award winning entrepeneur from Halden, that have reached high popularity with her cookie shop that went vial on TikTok. Jessica decided to not pursue higher education and rather go all in on the business. Jessica is single and activley searching. The business is going very well, in terms of sales, but her employees are struggling to plan the production of cookies as they rely on manual spreadsheets to track and plan production and inventory. Jessicas ambitions for her business is to grow exponentially by using her status as a TikTok star to expand to new markets.

Personas 3: Employee

Sara is 26 years old and works part-time in a bookstore in Tromsø while finishing her bachelor degree in media and communication. She is originally from Harstad and has worked in retail since high school. Sara is comfortable using modern apps, but she does not want to spend time learning complicated business systems during a busy shift. In her daily work, she mainly receives deliveries, adjusts stock, looks up items for customers, and reports damaged goods to the manager. Sara does not need full administrative control of the business, but she needs quick and reliable access to the parts of the inventory system that are relevant for her role.

Scenarios

Scenario 1: Ola

Ola have managed to secure a event with the famous author Jo Nesbø to sign copies of his latest book in the bookstore. Historically he know that such a event will lead to more sales, so he have to make sure he have enough copies in storage. Instead of going to the remote storage to count each book he want to be able to do a quick search to get the current stock to better be able to order more. He know he have sold many copies recently and know that the inventory is always updated in the app when he sells one.

Scenario 2: Jessica

Jessica have a idea to make a PopUp Store to sell her cookies in a new geographical location. She want to use this experiment to make content for her SoMe channels. Last time they did this it failed miserably, they ran

out of stock very fast and had to close the pop up store early due to lack of products. They did not know how much cookies they had at the central factory and the logistics behind the operation was a mess. The advantage of this system is that the sales staff get a notification when the stock gets low.

Scenario 3: Ola

Ola have several employees that work in different sections of the store. Since Ola will during 2026 be in maternitiy leave he have to make sure that the business runs optimal when he is away. He need the staff to be able to add books to the inventory when they get shipments. The staff also need to be able to do a simple lookup based on name, author, ISBN and genere. A additional feature is to report damaged books that is taken out of inventory. Only the admin and book store manager should be able to order goods when the inventory is low.

Scenario 4: Jessica

Jessica is in need to track inventory and sales easily. When inventory is low due to high sales they also need to be notified well in advance that they need to increase production of popular cookies before the inventory empty. A notification or warning to the production staff when inventory is below a certain threshold will make the production on demand based and self adjusting.

Scenario 5: Sara

Sara has just been invited to the bookstore's inventory system by Ola, the owner. After creating her account and logging in, she can see the bookstore in her list of inventories and selects it as her active inventory before starting her shift. During the day, she receives a small delivery of books and updates the stock directly in the system. Later, she discovers that some books have been damaged during handling and adjusts the stock so that the inventory remains correct. She can search for books and adjust stock, but she cannot change more sensitive settings or remove items completely, because those actions are restricted to the owner role.

Scenario 6: Ola

Ola now manages more than one inventory context and wants a quick overview before opening the store. After logging in, he selects the correct active inventory and opens the dashboard. There he can immediately see summary information and low-stock warnings, which helps him identify which books need attention first. When he notices that the stock of a popular title is lower than expected, he can open the item page and inspect the stock history to see what changes were made and by whom. This helps him understand whether the low stock is due to normal activity, a recent delivery update, or another stock adjustment. From there, he can quickly decide whether more books should be ordered or whether no immediate action is needed.

Features

Authentication and Account Management

Users can create accounts, log in, log out, verify sessions, and reset passwords so that access to the system is secure and user-specific.

Inventory Registration and Active Inventory Selection

Users can register a business, view the inventories they belong to, and select an active inventory so that all later actions happen in the correct business context.

Item and Stock Management

Users can create, edit, search, and delete items, adjust stock up or down, and view stock history so that inventory data stays correct and traceable over time.

Members and Invitations

Owners can invite employees, assign roles, and remove access so that inventory work can be shared while still enforcing role-based restrictions.

Dashboard and Low-Stock Monitoring

Users can view inventory summaries, low-stock warnings, and other quick insights so that they can identify important issues and act quickly.

Cross-cutting / Platform / UX

The system also includes supporting concerns such as navigation, guards, HTTPS, error pages, responsiveness, theming, and other platform-level UX improvements that affect multiple features.

Future / Integrations

Some backlog items are planned as future extensions, such as import/export, third-party integrations, and external API access.

Team Chat

A separate collaboration feature is planned for chat-based communication between team members, including item-related discussions when relevant.

User Stories

The list below contains the user stories from the InventoryX backlog. Full details for each user story, including descriptions, status, labels, and related GitHub issue information, are available in the complete backlog: InventoryX user story backlog.

1. As a business owner, I can log in so that I can access InventoryX securely
2. As a business owner, I can create an account so that I can access InventoryX securely.
3. As a business owner, I can increase or decrease an item's stock so that inventory matches sales, deliveries, and corrections.
4. As a business owner, I can add an item with a name and initial stock so that I can start tracking inventory quickly.
5. As a business owner, I cannot set stock to a negative number so that inventory data stays trustworthy.
6. As a business owner, I can see a list of all items with current stock so that I get a quick snapshot of inventory.
7. As a business owner, I can create register a business so that I can manage inventory for my company.
8. As a business owner, I can log out so that my account stays secure when I'm done.
9. As a User i cannot access pages without being logged in
10. As an employee, I can adjust stock levels so that inventory stays updated during my shift.
11. As a business owner, I can edit an item's details so that the inventory information stays correct when something changes.
12. As a business owner, I can search for items by name so that I can find products quickly.
13. As a business owner, I can register that new stock arrived so that inventory updates without manual recounting.
14. As a business owner, I can sort and filter items (e.g., lowest stock first) so that I can prioritize actions quickly.
15. As a business owner, I can see key inventory summary metrics so that I understand inventory status quickly
16. As a business owner, I can assign roles so that employees can update stock but not change sensitive settings.
17. As a business owner, I can invite an employee to join my business so that they can help update inventory.
18. As an business owner, I can only see and manage items within a selected inventory so that my data is separated from other businesses.

19. As a user, I can view my inventories so that I can choose which business to work with.
20. As a user, I can select an active inventory so that the dashboard shows the correct business context.
21. As a business owner, I can navigate between inventory, item details, and employee management so that I can complete tasks efficiently.
22. As a visitor, I can see a landing page so that I understand what InventoryX is and how to get started.**
23. As a user, I can see the correct website name and icon in the browser tab so that I can easily identify the Inventory X website.
24. As a business owner, I can reset my password so that I can regain access if I forget it.
25. As a user, I can log in with a social account so that I can access the app faster
26. As a user, when I want to edit or add items, I want the input fields to dynamically update to match exactly what the system currently requires, so that I only see relevant options and provide the right information without confusion.
27. As a business owner, I can delete an item so that I only track products I currently sell.
28. As a business owner, I can search by category so that I can find items even when names are similar
29. As a business owner, I can see a history of any changes to items so that I understand what changed and when.
30. As a business owner, I can upload an image to an item so that I can identify products visually.
31. As an Owner, I want to define custom fields for my inventory so that I can track item details specific to my business needs alongside the standard system fields
32. As a business owner, I can open an item page to see full details so that I can review information without scanning the full list.
33. As a business owner, I can set a low-stock threshold for an item so that I get warned before it runs out.
34. As a business owner, I can see a low-stock warnings list so that I know what to reorder or produce more of.
35. As a business owner, I can receive an email when an item's stock changes from above the low-stock threshold to equal to or below it, if low-stock notifications are enabled for that item.
36. As a business owner, I can view simple charts (low stock, total value) so that I can spot trends quickly.
37. As a business owner, I can remove an employee's access so that only authorized users can manage inventory.
38. As a business owner, I can export inventory so that I can share it externally.
39. As a business owner, I can import items from a template so that onboarding is faster than manual entry.
40. As a business owner, I can see clear messages when inventory is empty or a search returns nothing so that I know what to do next.
41. As a business owner, I can access the latest version of the inventory system after updates so that my employees can use new fixes and features without service interruption.
42. As a employee, I can use the inventory system over a secure HTTPS connection so that login and session data are protected in transit
43. As a user, I can switch between light mode and dark mode so that the app is more comfortable to use
44. As a user i want to have all features available on small devices so i can access it from my phone

45. As an employee, I can choose a reason (sale/delivery/damaged/other) when updating stock so that updates are understandable later.
46. As a Owner, i can specify what unit the item has (L, Kg, packages, ect.)
47. As a Owner, i can do granular permission management for employees (if they are allowed to delete items, edit item info, see logs(?))
48. As a business owner, i could connect Vipps API and have transactions automatically decrease stock
49. As a Client, I want to access my project inventory via a REST API, so that I can programmatically manage stock levels and integrate with third-party tools.
50. As a business owner, I can create a chat room so that I can communicate with my team about inventory tasks.
51. As a business owner, I can send and view messages in a chat room so that my team can communicate about inventory work.**
52. As a business owner, I can view and send messages linked to a specific item so that product-related communication stays organized.
53. As a visitor, I can choose whether to accept non-essential cookies so that I can control my privacy preferences.
54. As a visitor, I can see clear error pages for unauthorized, forbidden, and missing pages so that I understand what went wrong and what I can do next.
55. As an employee, I can receive an email notification when I receive a new invite to an inventory, So that I know exactly when I have been granted access and can act on it.
56. As a user, I can see the current inventory and access Items from the navigation bar so that I always know where I am working and can navigate faster
57. As a user, I can read a privacy policy so that I understand what personal data is collected and why.